

DIABETIC RETINOPATHY SCREENING REPORT

Smart India Hackathon 2026 | Problem 26038 | MathWorks AI Pipeline

Image: download

Date: 09-Sep-2026 09:49

Quality: BORDERLINE (2/3)

1 AI DIAGNOSIS

DIAGNOSIS

Proliferative DR

ICDR Severity Level 4 / 4 Confidence 57.4%



URGENT REFERRAL — Ophthalmologist consult within 7 days

2 IMAGE QUALITY METRICS

Sharpness (Laplacian Variance) Threshold: ≥ 50	1.8	FAIL
Brightness (Mean Intensity) Threshold: 30 – 210	131.8	PASS
Contrast (Std Deviation) Threshold: ≥ 25	29.6	PASS

3 DETECTED LESION EVIDENCE

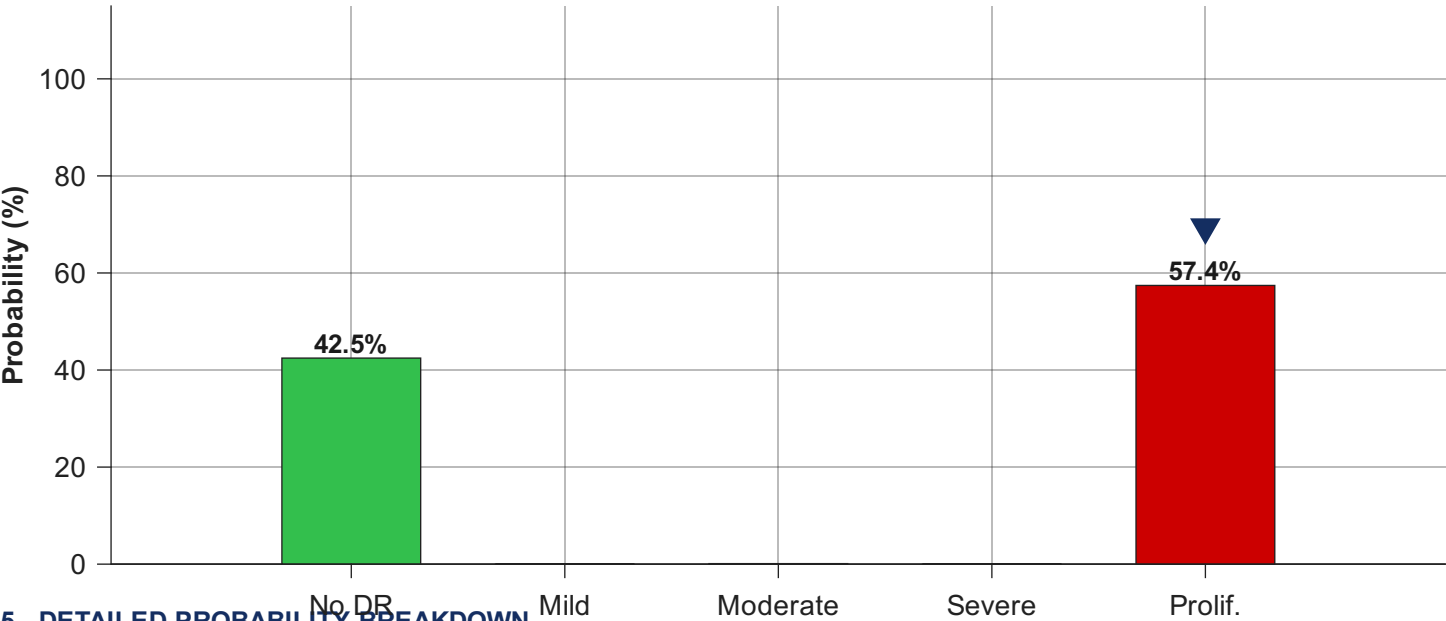
Microaneurysms (MA) Earliest DR sign — small vascular microleaks	0 detected
Hard Exudates (EX) Lipid deposits — blood-retinal barrier breakdown	48 detected
Hemorrhages (HE) Deeper vascular wall ruptures	7 detected

Optic Disc [476, 227] Fovea [393, 232]

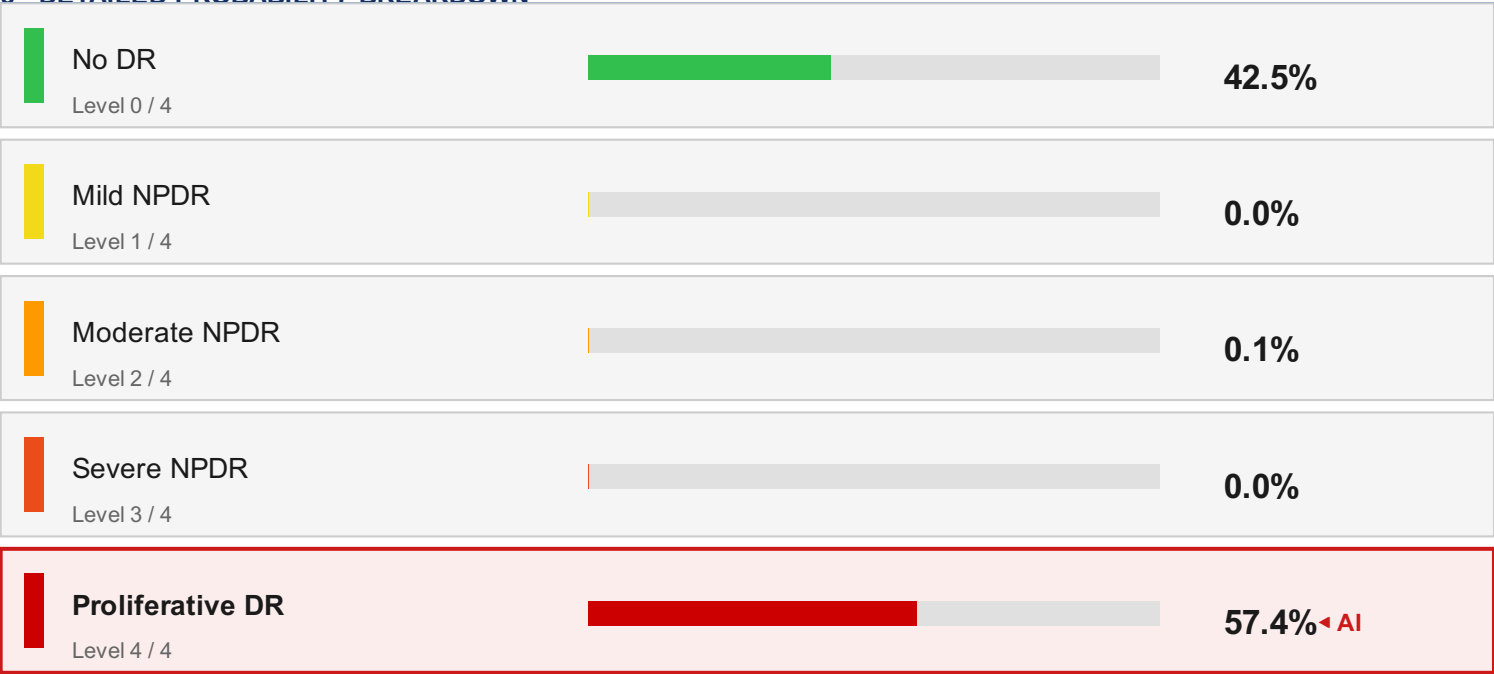
CLASS PROBABILITY ANALYSIS

Smart India Hackathon 2026 | Problem 26038 | MathWorks

4 PROBABILITY DISTRIBUTION



5 DETAILED PROBABILITY BREAKDOWN



FUNDUS IMAGE QUALITY & ENHANCEMENT

Smart India Hackathon 2026 | Problem 26038 | MathWorks

A ORIGINAL FUNDUS IMAGE



B CLAHE ENHANCED & ILLUMINATION NORMALISED



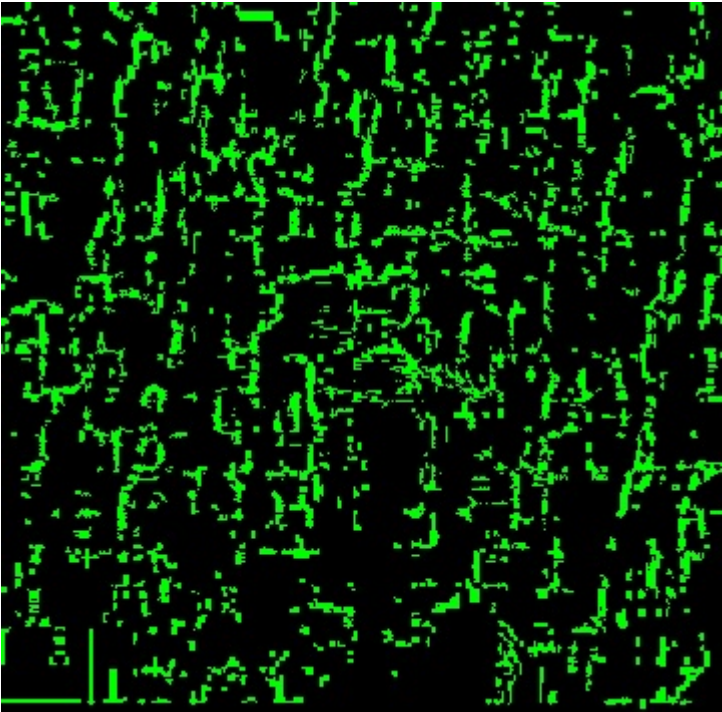
RETINAL SEGMENTATION & EXPLAINABLE AI

Smart India Hackathon 2026 | Problem 26038 | MathWorks

C. Anatomical Landmarks (OD=Yellow, Fovea=Cyan)



D. Vessel Network Segmentation



E. Lesion Map (MA=0, EX=48, HE=7)



F. Grad-CAM Explainability (57.4% Confidence)

